

SAFETY PRECAUTIONS

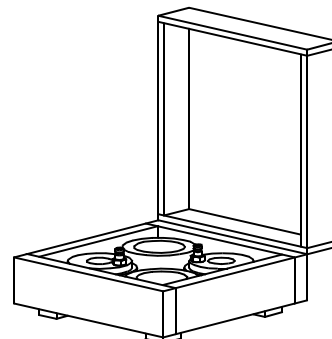
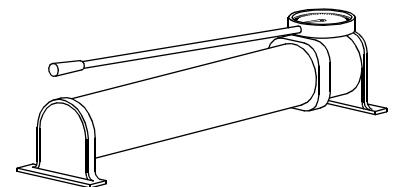
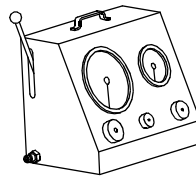
☐	Stopped engine
☐	Block the starting mechanism
☐	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D-1	Tightening pressure for holding down bolts and end chock bolts	1500	bar
D-2	Check holding down bolts and end chock bolts for correct tightening after the following service hours:		
	1st check	500	hours
	2nd check	1000	hours
	3rd check	2000	hours
	4th check	4000	hours
	5th check	8000	hours
	Subsequently at intervals of	8000	hours

Engine on **Epoxy** supporting chocks

Standard Tools: See Section 913





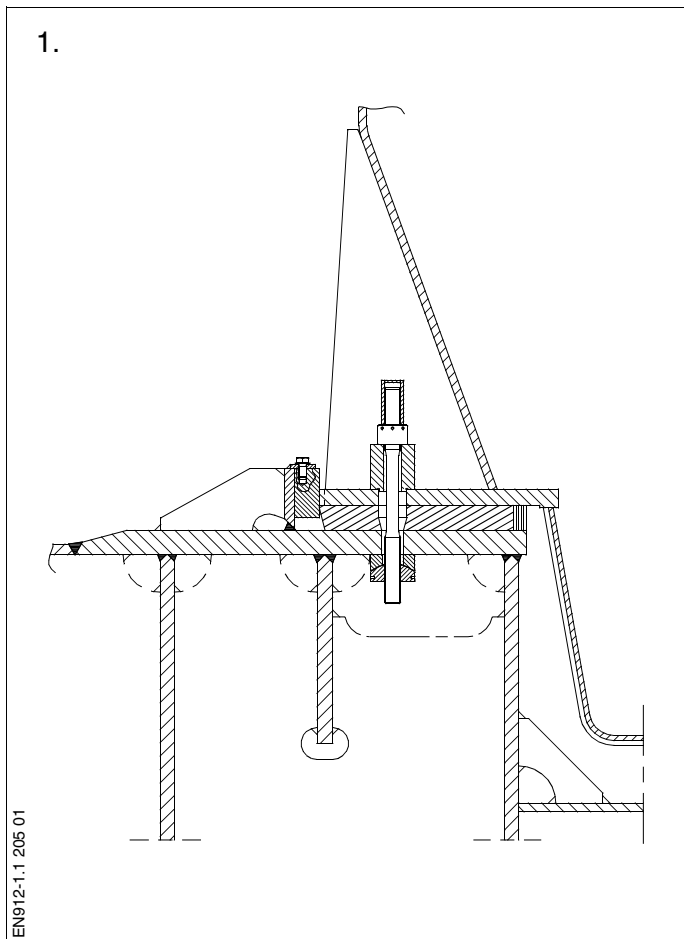
Spare Parts

Plate – Item No.	Description	Qty
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Note!
The hydraulic jack used for tightening the holding down bolts is marked with: For: Holding down bolts The larger jack used for tightening the end chock bolts is marked with: For: End chock bolts

Hydraulic tightening of holding down bolts and end chock bolts is carried out as detailed in Section 913-1. The normal tightening pressure is indicated on the Data sheet and is also stamped on the tightening tool.

1.



Checking the Bolt Tightening

1. The holding down bolts and end chock bolts must be checked for correct tightness at the intervals indicated on the Data sheet.

For this purpose, raise the pressure on the hydraulic tool slowly while constantly attempting to loosen the nut with the tommy bar. The oil pressure indicated on the pressure gauge when the nut comes loose ('loosening pressure') is to be noted down in the checking tables, see pages 4 and 5, following which the bolts are tightened to the normal tightening pressure.

The condition of the bolted joints, and thus the general condition of the foundation, can be effectively checked by comparing the tables from successive bolt checks.

If the 'loosening pressure' is below 80 per cent of the tightening pressure, the relative chocks shall always be checked for possible defects. If the chocks are in position and in order, the bolts should be taken out for inspection of threads and contact faces.

Checking of Epoxy Supporting Chocks

2. If a number of measuring pins have been welded to the tanktop, the heights of the epoxy supporting chocks are to be checked immediately after finishing the checking of the loosening pressures of the holding down bolts and the retightening of these.

The distance between the measuring pins and the bedplate is to be measured with a blade gauge and noted down. Any possible settling of the chocks during the intervals between measurements can thereby be followed.

Side Chocks and Side Chock Liners

3. After fitting the liners to an 80 per cent contact area on both sides of the liners, knock the liners a further 3 to 4 mm inwards.

The liners located in way of each main bearing on either side of the engine must be fitted and knocked into position simultaneously.

The first time the ship is sailing in a fully-loaded condition after the engine has been operating for 1,000 hours, all side chocks should be checked to see whether the liners can be knocked further inward.

The fit of the side chock liners should be checked with a feeler gauge each time the loosening pressure of the holding down bolts is checked, and thus at the same time intervals.

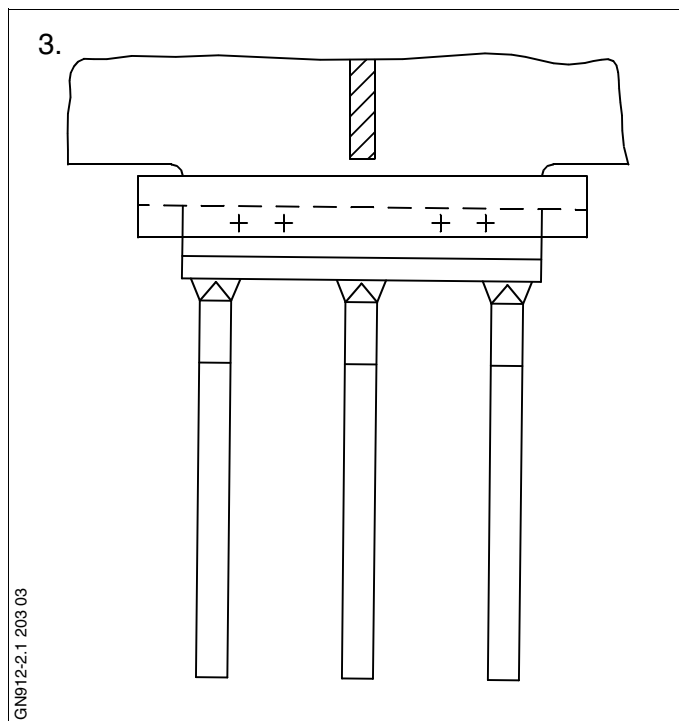
The feeler gauge is applied at the 7 points indicated in the table on page 5, and the measurements found are to be entered in the relevant table. These results are used to determine whether refitting or, possibly, replacement of the liners is necessary.

If the measurements at 3 points or more have increased 5/100 mm or more from the initial results, we recommend that the following procedure is followed:

1. Loosen the hexagon screws.
2. Try to knock the liner further inward.
3. Measure again at the 7 points indicated in the table.

If this procedure does not improve the situation, the liner must be removed, and it must be checked that the actual contact area is more than 80 per cent of the possible contact surface areas on both sides of the liner.

The liners are secured in their correct position by means of hexagon socket set screws with cup point.



GN912-2.1 203 03

Holding Down Bolts

Engine frame No.	Bolt No.	Port side			Cylinder No.	Starboard side			Bolt No.	Engine frame No.
		pressure	% dev.	remarks		pressure	% dev.	remarks		
1	1				1				1	1
	2								2	
	3								3	
	4								4	
2	1				2				1	2
	2								2	
	3								3	
	4								4	
3	1				3				1	3
	2								2	
	3								3	
	4								4	
4	1								1	4
	2								2	
	3								3	
	4								4	
5	1								1	5
	2								2	
	3								3	
	4								4	
6	1								1	6
	2								2	
	3								3	
	4								4	
7	1								1	7
	2								2	
	3								3	
	4								4	
8	1								1	8
	2								2	
	3								3	
	4								4	
9	1								1	9
	2								2	
	3								3	
	4								4	
10	1								1	10
	2								2	
	3								3	
	4								4	
11	1								1	11
	2								2	
	3								3	
	4								4	
12	1								1	12
	2								2	
	3								3	
	4								4	
13	1								1	13
	2								2	
	3								3	
	4								4	
14	1								1	14
	2								2	
	3								3	
	4								4	
	1								1	

Side Chocks and End Chock Bolts

Port side								Starboard side										
Chock No.	Point							Cylinder No.	Point							Chock No.		
	A	B	c	d	e	f	g		A	B	c	d	e	f	g			
1								1 2 3								1		
2																		2
3																		3
4																		4
5																		5
6																		6
7																		7
8																		8
9																		9
10																		10
11																		11
12																		12
13																		13
14																		14

Port side

FORE

Starboard side

Port side					Starboard side								
Ch No.	Bolt No.	Pressure (kp/cm ²) bar	Deviation %	Point					Deviation %	Pressure (kp/cm ²) bar	Bolt No.	Ch No.	
				a	b	c	d	e					a
E	1											1	E
	2											2	

Port side

AFT

Starboard side

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D12-04	Check measurement	148±3	mm
D13-01	Hydraulic pressure, mounting	1500	bar
D13-02	Hydraulic pressure, dismantling	1400-1650	bar

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P91261		Large Parts - Tools
P91351	010	Hydraulic pump, pneumatically operated
P91351	046	Hose with unions (1500 mm), complete
P91351	058	Hose with unions (3000 mm), complete
P91351	105	3-way distributor block, complete

1. This check can be performed during several periods of stopped engine, but must be performed on complete twin stay bolt groups at a time, thereby ensuring uniform tension in the affected bearings.

Note!

The first time this check is performed it has to be completed in one operation. The following checks have to be completed within a 500-hour period.

2. Before retightening the stay bolts, remove the protective caps.
3. Check for loose staybolt nuts with the tommy bar.

If a loose nut is found, unscrew the nut until a gap occurs between cylinder frame and nut.

Loosen and unscrew the nut on the opposite stay bolt.

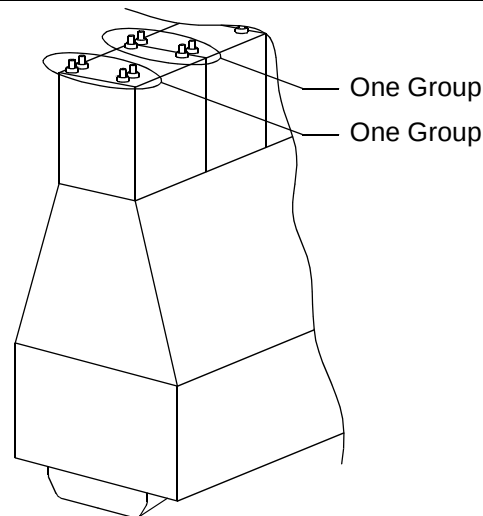
Check for the correct mounting of the stay bolt by comparing with the check measurement on the Data Sheet.

If the check measurement is not reached, indicating that the stay bolt is not screwed properly down in the thread of the bedplate, apply a screw and a conternut in the thread in the top of the stay bolt, and turn the stay bolt until the check measurement data is met.

4. Retighten the stay bolts in pairs (athwartship), working from fore to aft of the engine.

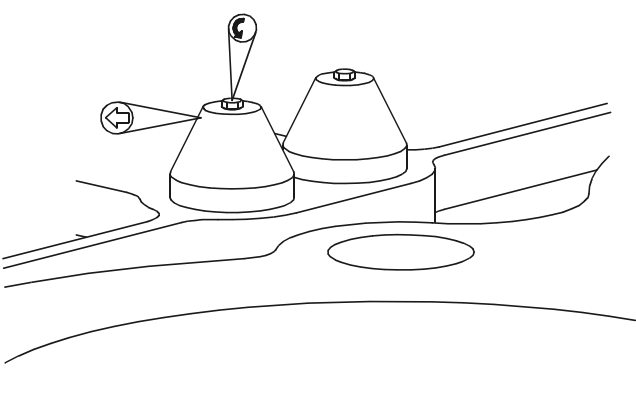
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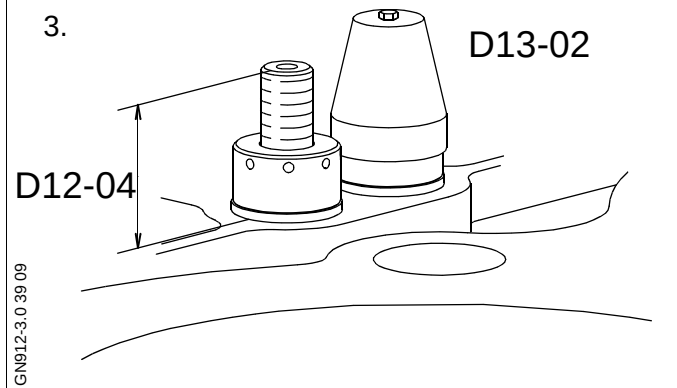
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3.

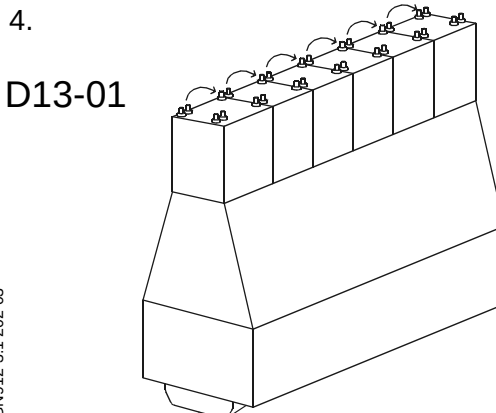
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4.

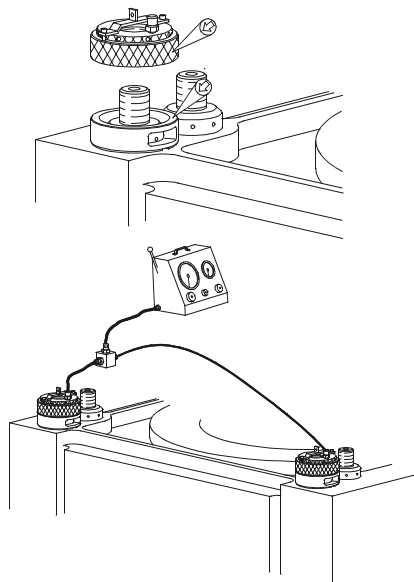
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D13-01



5.

D13-01



5. Clean the contact faces and mount the hydraulic tools on a pair of stay bolts positioned opposite each other.

Connect the high-pressure pump by means of the high-pressure hoses, so that the two stay bolts are tightened in one operation.

6. Maintain the mounting pressure, and retighten the stay bolt nuts with a tommy bar. Before relieving the system of pressure, check with a feeler gauge that the nuts bear against their contact faces.
7. When all stay bolts have been retightened, reconnect the hydraulic tools to the pair of stay bolts first tightened. Tighten to 10% below the mounting pressure and check whether the nut is loose.

If the nut is **not** loose:

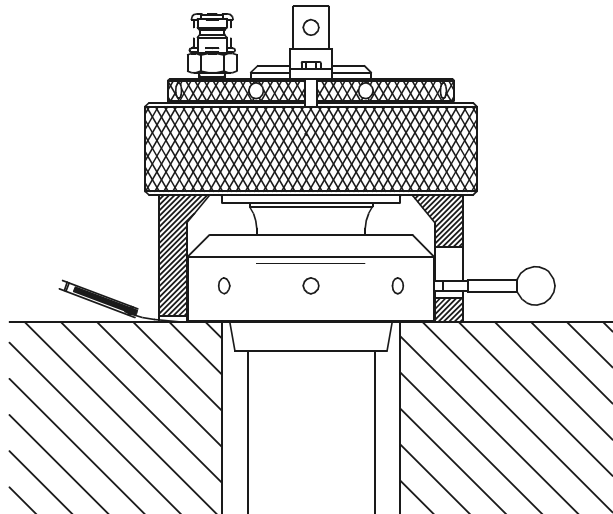
Tighten the stay bolts concerned to the mounting pressure.

If the nut is loose:

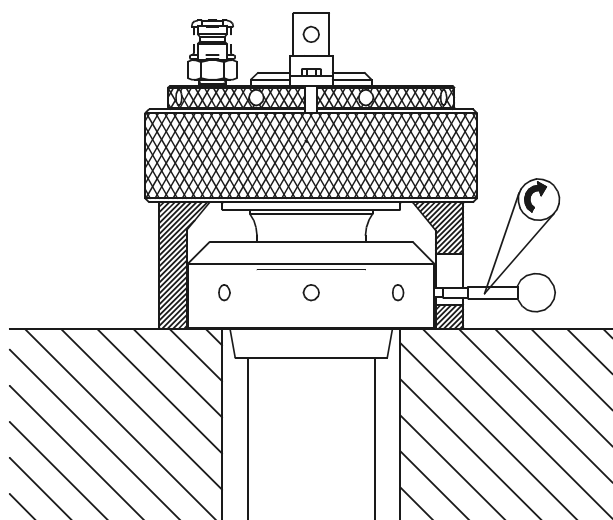
Tighten **all** stay bolts once again to the mounting pressure.

8. After completing the retightening procedure, mount the protective caps.

6.



7.



The crankcase oil outlets guide the lubricating oil from the crankcase to the lubricating oil bottom tank. The sealings of the crankcase oil outlets must be checked at regular intervals, for example during dockings. The crankcase oil outlets may be equipped with either rubber diaphragm sealing or metal bellow sealing.

Note!

If the water content of the main engine lube oil is rising, this may indicate that the crankcase oil outlet sealings are fractured.

Rubber diaphragm sealing

- To access the rubber diaphragm sealing remove:
Screws A
Grating B
Screws C
Cover plate D
Screws E
Steel ring F.
- Lift away the rubber sealing diaphragms G and examine each diaphragm closely. In case of any rips or tears in the diaphragms, they must be replaced.

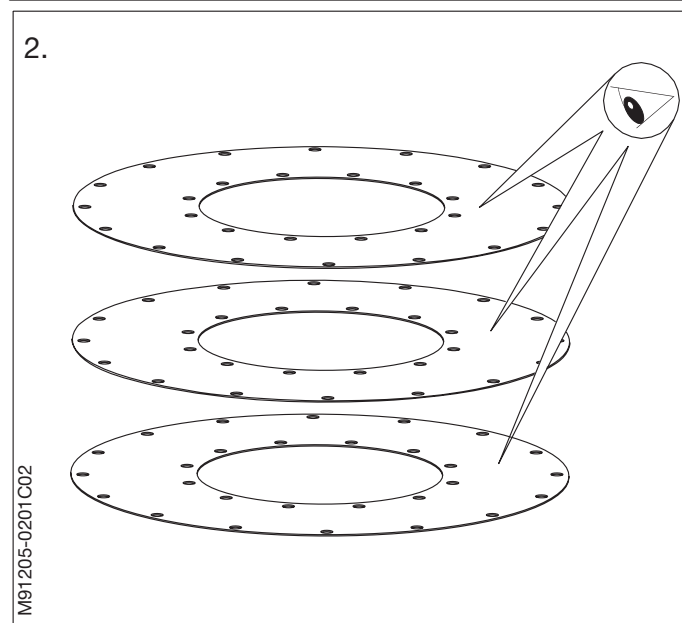
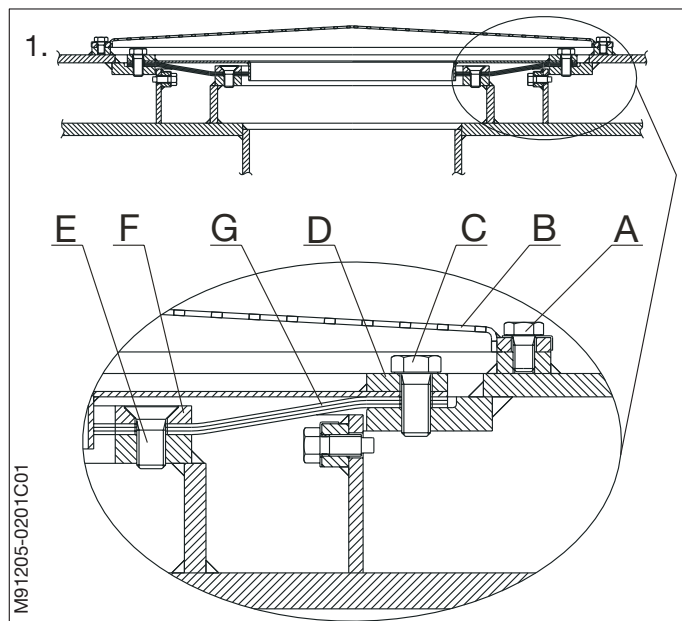
Note!

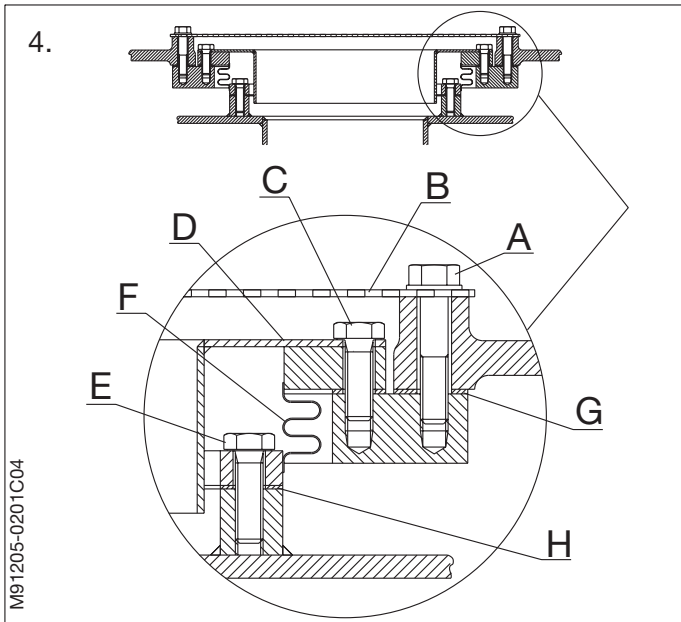
It is strongly recommended to always replace the diaphragms during inspection. If unavailable, new diaphragms may be made from three layers of 2 mm thick oil and temperature resistant rubber.

- Mount:
Rubber sealing diaphragms G
Steel ring F
Screws E
Cover plate D
Screws C
Grating B
Screws A.

Note!

Remember to fit new locking plates at screws A.

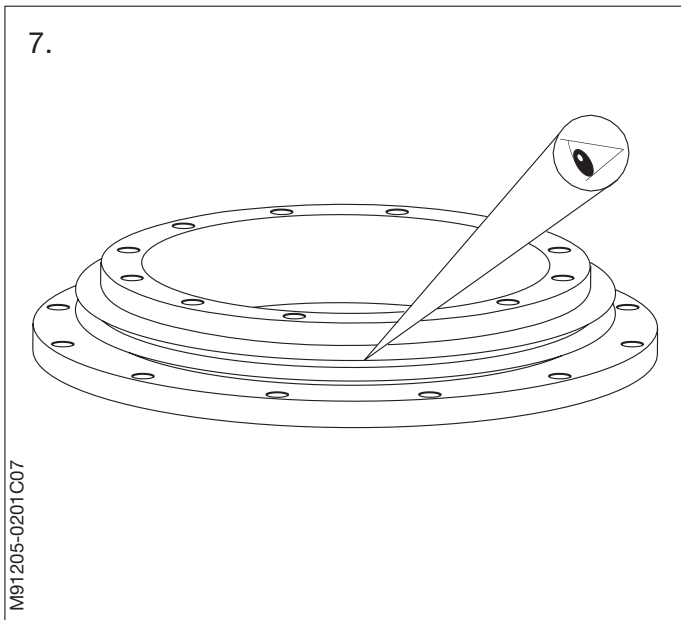


**Metal bellow sealing**

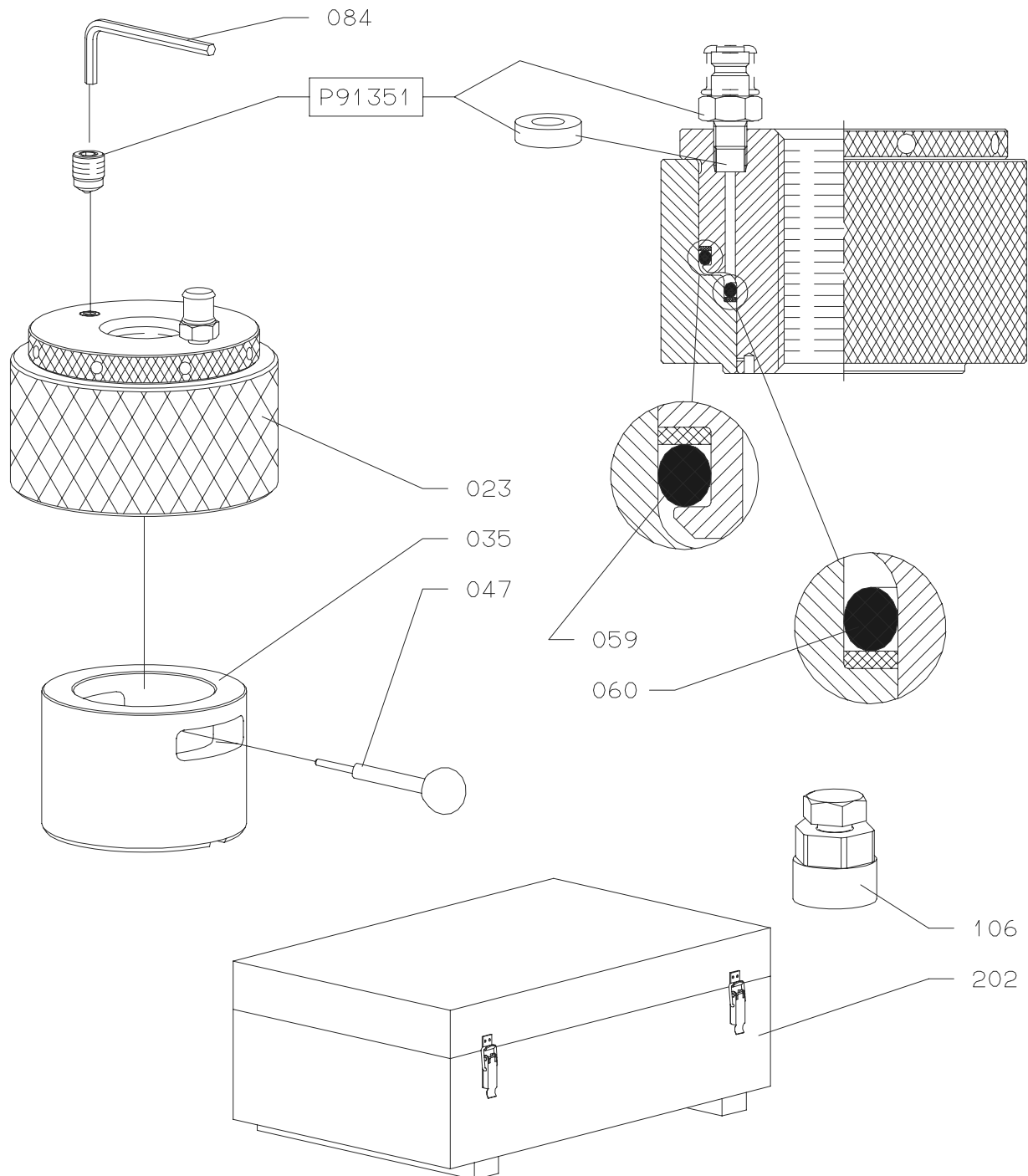
4. Remove all screws A and grating B.
5. Remount four of the screws A at diametrically opposite positions.
6. Remove:
Screws C
Cover plate D
Screws E.
7. Lift away metal bellow sealing F and examine it closely. If any cracks or punctures are found in the metal bellow sealing, it must be replaced.

Note!

It is recommended to always replace the metal bellow sealing during inspection.



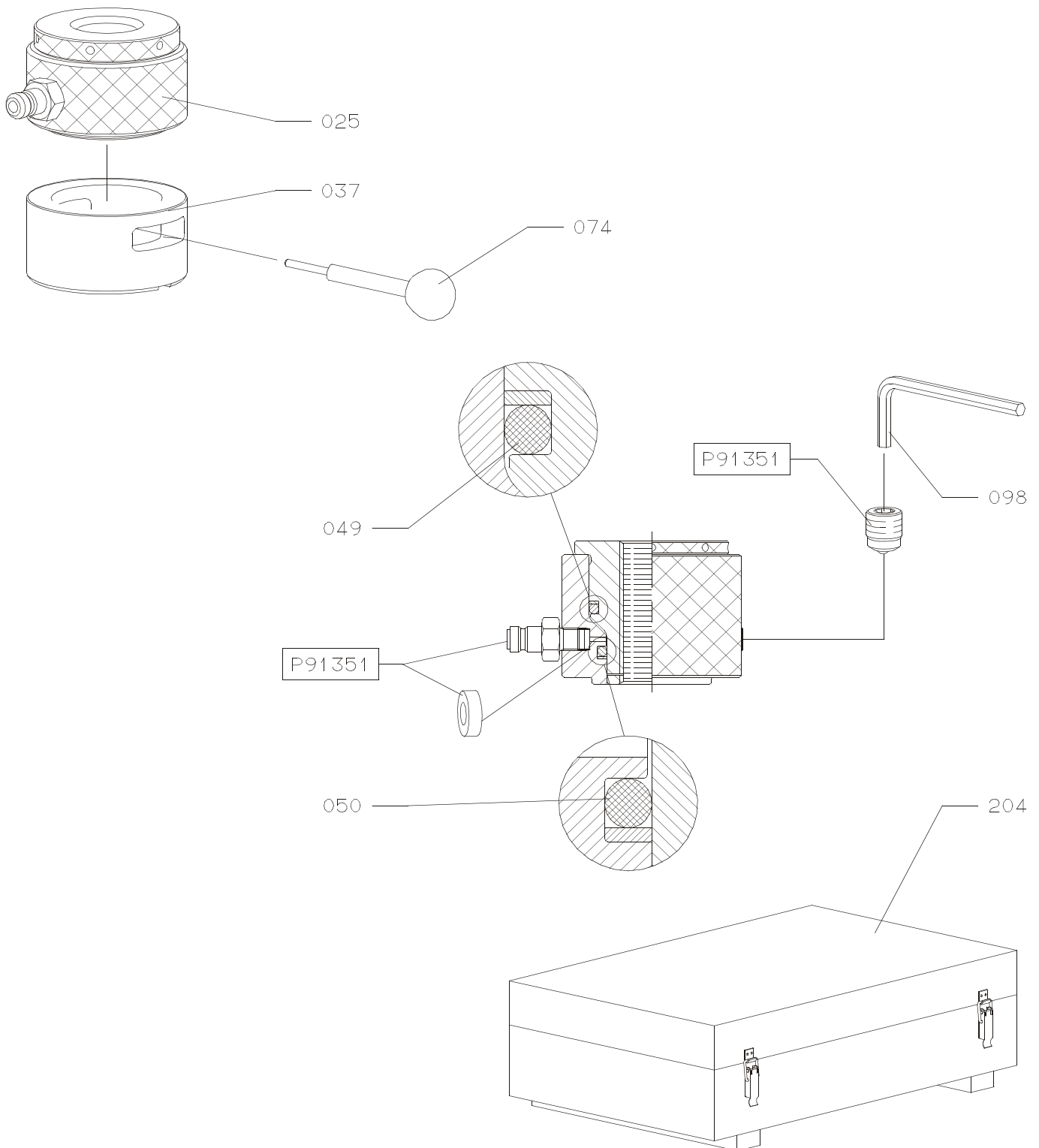
8. Replace gaskets G and H.
9. Mount metal bellow sealing F.
10. Mount:
Screws E
Cover plate D
Screws C.
11. Remove the four screws A.
12. Mount grating B.
13. Mount all screws A.





Item No.	Item Description
023	Hydraulic jack, complete
035	Support
047	Tommy bar
050	Sealing ring with back-up ring
060	Sealing ring with back-up ring
084	Spanner
106	Stud setter
202	Hydraulic tool set, complete

Item No.	Item Description

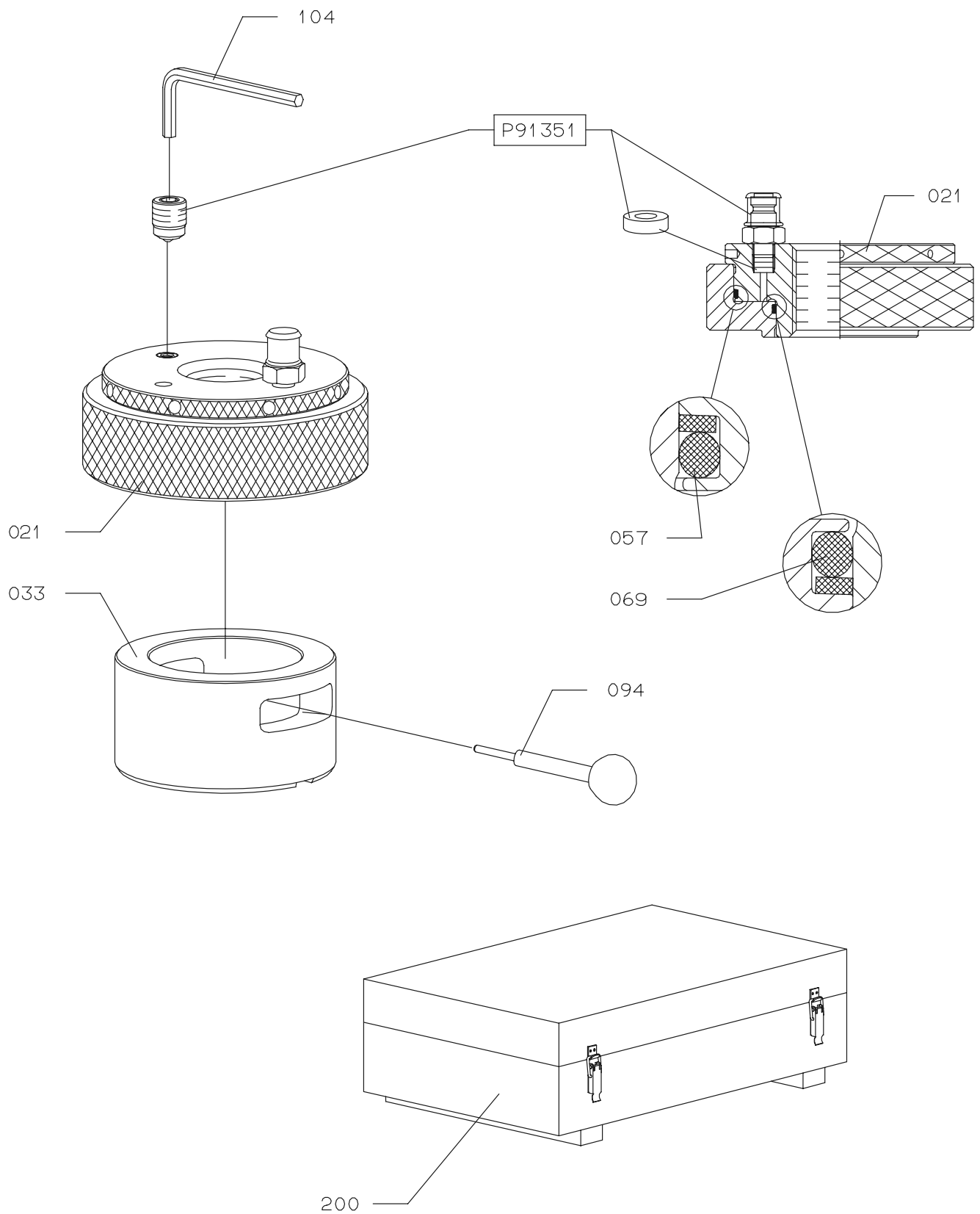


Hydraulic Tools - Holding Down Bolts (Epoxy Chocks)



Item No.	Item Description
025	Hydraulic jack, complete
037	Support
049	O-ring with back-up ring
050	O-ring with back-up ring
074	Tommy bar
098	Key, hexagon socket screw
204	Hydraulic tool set, complete

Item No.	Item Description





Item No.	Item Description
021	Hydraulic jack, complete
033	Support
057	O-ring with back-up ring
069	O-ring with back-up ring
094	Tommy bar
104	Spanner
200	Hydraulic toolset, complete

Item No.	Item Description